



### Wearable Extra thin LED

University of Tokyo researches have invented extra thin LED's (3 micrometers) which blend with your skin. <http://dgit.in/SknDisp>



### Customise Your Nexus Case

Nexus users can now customise and make their own smartphone cases using the Play Store. <http://dgit.in/Nxscase>



Drones can now actually "see" thanks to VPUs

Qualcomm, the biggest name in smartphone SoCs, recently introduced a bunch of enhancements made to its chips using specialised algorithms. In its data centres, Microsoft uses a specialised FPGA (Field-Programmable Gate Array) chip for Bing. The company told The Economist that it has doubled the number of queries a server can process in a given time.

Given the DJI Phantom 4's proficiency in obstacle avoidance and Google's recent showcase of the Project Tango devices, it seems like the industry has overcome this hurdle as well.

Lastly, Dashwood says that VPUs take almost no discernible space, which makes it easier to implement into smaller devices such as smartphones and smartwatches.

"The additional sensors often involved are much larger considerations when it comes to space," he says.

### Application in virtual reality

While the implementation of VPUs in augmented reality is apparent, the industry today has been primarily focusing its attention on virtual reality. For starters, a VPU can help make Virtual Reality headsets less bulky, says Dashwood. More importantly, it can add a ton of ways for Virtual Reality applications to interact with the real world. Room-scaling in HTC's Vive headset is one such example of how virtual reality can work in conjunction with the real world. Think of it as a union between Virtual and Aug-

mented Reality. What if the virtual space was built around your real space?

"VPUs can help in 'all sorts of areas'", says Dashwood. He lists a positional tracking, gesture, environment mapping, eye tracking and object classification as a few examples. These are some of the essential components of Virtual Reality today. If your environment can be effectively mapped, then the virtual reality space a headset like the Oculus Rift takes you to, can be built around it.

This means if you're in your living room, your Minecraft game will be built based on things in it. Thinking back to the legendary Age of Empires games, imagine players in a single house, building their empires in separate rooms of the house, while the doorways act as borders between their empires.

Microsoft's Hololens is another area where VPUs can come in handy. The augmented reality headset seems to be one of the best things to have come out of Microsoft's stables recently, and it essentially depends on recognising the real world and then overlaying the virtual on top of it.

VPUs are not just possible for VR, they're almost essential.

### Application in smartphones

VR though is still about a year or so away from truly coming to the mainstream and consumers today are still more focused on smartphones. A very interesting possibility for VPUs in smartphones is in improving cameras on them. "Computational photography is an obvious application," says Dashwood, "there are a great deal of ways of working around the physical limitations of optics running on various operations, to construct a visually pleasing photograph." According to him, computational photography has the potential to bring DSLR (or better) quality images to our smartphones.

In essence, Computer Vision allows your smartphone to understand the scene in front of you. So, a photograph clicked by the camera, which is based on the reflection and refraction of light waves from various objects, can be improved. A photograph can be passed through additional processing, adding inputs from the VPU, to generate more realistic representations.



Don't try this at home, or at least not without a safety drone